Multi-Modal Scientific Figure Understanding

Project Status Report — what is done and what is not

Branch **claude/scientific-figure-understanding-9i8ve7** · generated 2026-10-14 from **results/*.json**

Status label	Meaning
DONE	Implemented, tested and measured.
LIMITED	Works end to end, but simplistic or weak in measured quality.
NOT DONE	Not implemented or not evaluated yet.
BLOCKED	Cannot be done in the current cloud environment (network policy).

All numbers in this report are read automatically from the evaluation result files. Numbers reported before the 2026-10-08 audit were produced by hard-coded constants and have been removed; they must not be cited.

1. Executive Summary

The project builds a system that reads a scientific paper and its figures, extracts the text in each figure, finds scientific entities (methods, metrics, datasets, variables), links figure elements to those entities in a graph, refines the links with a graph neural network (GNN), handles 3D figures with dedicated encoders, retrieves figures from text, and generates and verifies captions. The full pipeline runs end to end as scripts, a FastAPI backend and a React user interface, with automated tests and continuous integration on GitHub.

It was evaluated on a 400-figure controlled benchmark (with real protein structures for the 3D figures) and on **287 real arXiv figures from SciCap**. The key outcomes are:

- Entity grounding: the learned model improves link F1 from 0.38 (rules) to 0.91 on the synthetic benchmark — but the same model without message passing scores 0.91, so the gain comes from the learned features, not from the graph.
- Real figures: caption-to-figure retrieval reaches MRR 0.897; adding the figure's OCR text gives a small, statistically significant gain (+0.021 [+0.003, +0.041] ✱).
- Adding 3D or full multimodal information has **no detectable effect** on grounding, retrieval or caption quality (all confidence intervals contain zero); 3D adds a new capability (finding the figure that shows a given 3D structure).
- The 3D image classifier does not transfer to real papers (it flags 92.7% of real 2D graph plots as 3D); the default now requires image and text to agree, which reduces this to 2.4%.
- Weak points: generated captions are poor (BLEU-4 0.051 against author captions), entity extraction finds nothing in 130 of 287 real figures, and grounding accuracy on real figures cannot be measured yet because no labelled data exists.

2. System Overview

Stage	What it does	Implementation
Document ingestion	PDF → sections → paragraphs	PyMuPDF
Scientific NER	Finds methods, metrics, datasets, variables	Regex vocabulary (prototype)
Hierarchical retrieval	Paper / section / paragraph / entity context	FAISS + Sentence-Transformers; word-overlap fallback
Figure processing	Reads axis labels, legends, labels	EasyOCR + layout rules
Grounding graph	Links figure elements to entities	NetworkX rules + supervised PyG GAT model
3D branch	Detects 3D figures; encodes 3D geometry	CNN detector, PointNet, multi-view CNN
Multimodal retrieval	Finds figures from text	Fusion of text / OCR / graph / 3D channels
Caption + verification	Writes a caption and checks its claims	Evidence template + lexical/numeric checker
SCQ metric	Composite caption quality score	Weighted sum of four measured components
Interfaces	Run and inspect results	FastAPI backend, React 19 + Vite UI, CI

3. What Is Done

Components that are implemented, tested and backed by measurements.

3.1 Audit and clean-up

- Repository audit (docs/AUDIT.md) found that earlier benchmark numbers were hard-coded (e.g. GNN accuracy = baseline + 0.12) and that several components were placeholders.
- Fabricated result files were deleted; evaluation scripts were rewritten to measure everything they report; regression tests now fail if hard-coded metric offsets reappear.
- Placeholder data were relabelled (the 'AlphaFold' point cloud was a synthetic helix).

3.2 Datasets

Dataset	Content	Use
SciFig-Controlled (synthetic)	400 matplotlib figures (282 train / 118 test); exact labels for elements, entities and links; 3D molecular figures from 16 real PDB structures, split by structure	Training and controlled evaluation of all components
SciCap subset (real)	287 real arXiv graph plots with author captions and paper paragraphs (CC BY-NC-SA 4.0); fetched with scripts/fetch_scicap_subset.py	Evaluation on real papers

3.3 Figure element extraction (OCR)

Measure	Synthetic benchmark	Real SciCap (287 figures)
Text precision / recall / F1	0.82 / 0.80 / 0.81	not measurable (no gold boxes)
Element type accuracy	0.86	–
Text elements per figure	–	9.1 (19 figures with none)
Latency (CPU)	–	2.8 s per figure

3.4 Entity grounding (Phase 3 — GNN)

Synthetic test split, 3 training seeds (mean ± std). 'Oracle' uses the true figure text; 'OCR' uses text read from the image.

Method	F1 (oracle)	F1 (OCR, end to end)	F1 (unseen entity names)	Inference / figure
A. Rules (NetworkX)	0.397	0.381	0.418	0.12 ms
B. GNN (GAT + pair classifier)	0.995 ± 0.002	0.905 ± 0.003	0.833 ± 0.013	4.6 ms
Same model, no message passing	0.990 ± 0.003	0.910 ± 0.002	0.846 ± 0.025	2.5 ms

Finding: the learned model clearly beats the rules, but message passing itself adds nothing measurable. Training takes a few seconds on CPU; inference is a few milliseconds per figure (training and inference time are reported separately).

3.5 3D branch (Phase 4)

Component	Result	Notes
3D figure detector (synthetic test)	accuracy 1.00, figure-type accuracy 1.00	In-domain only
3D detector on real graph plots	false positives: image-only 92.7%, text-only 2.4%, agreement 2.4%	'Agreement' is now the default when text is available
Figure ↔ 3D structure retrieval (R@1)	PointNet 0.138 ± 0.015, multi-view 0.159 ± 0.026	chance 0.023, untrained 0.016; 3 seeds; unseen proteins
3D in the grounding graph	FIGURE_3D node with image-derived 3D embedding	Used in ablation D/E

3.6 Ablation study A–E (synthetic test split)

Config	Grounding F1	Caption→figure MRR	Geometry→figure R@1	SCQ
A. NetworkX rules	0.381	0.860	N/A	0.545
B. NetworkX + PyG GNN	0.905 ± 0.003	0.876	N/A	0.569
C. GNN + CLIP	BLOCKED — CLIP weights unavailable	–	–	–
D. GNN + 3D branch	0.909 ± 0.001	0.877	0.093	0.571
E. Full multimodal (text + 2D + 3D + GNN)	0.905 ± 0.003	0.881	0.093	0.572

Comparison (paired bootstrap, 95% CI)	Grounding F1	Retrieval MRR	SCQ
B − A: GNN vs rules	+0.525 [+0.500, +0.547] ✱	+0.016 [-0.009, +0.042]	+0.024 [+0.004, +0.044] ✱
D − B: add 3D	+0.003 [-0.005, +0.010]	+0.001 [-0.004, +0.006]	+0.002 [-0.000, +0.005]
E − B: full multimodal	+0.001 [-0.009, +0.010]	+0.004 [-0.020, +0.029]	+0.003 [-0.003, +0.008]

✱ = confidence interval excludes zero (statistically significant). Answer to the Phase 4 research question: on this benchmark, explicit 3D information does not measurably improve grounding, retrieval or SCQ; its value is enabling geometry-to-figure retrieval.

3.7 Real-figure evaluation (SciCap)

Retrieval channels	R@1	R@5	MRR	vs text only
text only	0.875	0.920	0.897	–
text+ocr	0.899	0.937	0.919	+0.021 [+0.003, +0.041] ✱
text+grounded	0.875	0.927	0.898	+0.000 [-0.005, +0.006]
text+ocr+grounded	0.899	0.937	0.918	+0.021 [+0.003, +0.039] ✱
ocr only	0.477	0.568	0.521	-0.376 [-0.435, -0.319] ✱

Data-leak control: 35 of the paper paragraphs contained the caption word for word, which would inflate retrieval (leaky MRR 0.945). Captions and 'Figure N:' labels are removed before measuring.

3.8 Engineering

- FastAPI backend: each sample runs on its own text; path-restricted figure-image endpoint; honest status fields.
- React UI: analysis view shows the real document, figure image, OCR boxes and grounding links; a warning banner appears whenever built-in demo data is displayed.
- Continuous integration on GitHub runs the test suite and the frontend build on every push.
- Automated tests (39) including regression tests for every issue found in the audit.
- Reproducibility: fixed seeds, multi-seed reporting, paired bootstrap confidence intervals, separate training and inference timings, and hardware recorded in every results file.

4. What Works but Is Limited

Area	Status	Limitation (measured)
PDF ingestion	**LIMITED**	Each text line becomes a 'paragraph'; figures are not extracted from PDFs.
Scientific NER	**LIMITED**	Regex over a fixed vocabulary; no entity found in 130 of 287 real figures.
Caption generation	**LIMITED**	Template, no vision-language model. Real figures: BLEU-4 0.051, ROUGE-L 0.084.
Claim verification	**LIMITED**	Word overlap + numeric conflict, no NLI model. Real-figure correctness 0.26.
SCQ on real figures	**LIMITED**	0.30 ± 0.16; mainly reflects the weak captions.
OCR quality	**LIMITED**	Real-figure text is often garbled (e.g. 'Accuracy Score' read as 'Accuangy Scine').
3D alignment	**LIMITED**	Clearly above chance but weak (R@1 ≈ 0.15); trained on 137 synthetic figures.
Synthetic benchmark	**LIMITED**	Template text and a single plotting style; results do not transfer automatically to real papers.

5. What Is Not Done

Item	Status	Reason / what is needed
Grounding accuracy on real figures	**NOT DONE**	SciCap has no element → entity labels; ~100 hand-annotated figures are needed.
Evaluation on real 3D figures	**NOT DONE**	SciCap contains graph plots only; no labelled real 3D figure set is available.
Cross-Modal Search tab (UI)	**NOT DONE**	Still shows hard-coded example results; not connected to the retrieval code.
Uploading your own PDF / figure in the UI	**NOT DONE**	The UI sends only file names, not file contents; it runs the bundled samples.
Figure extraction from PDFs	**NOT DONE**	Figures must be supplied as separate image files.
Useful GNN message passing	**NOT DONE**	Currently adds nothing measurable; needs redesign (e.g. cross-figure graphs) or removal.

6. What Is Blocked

Item	Status	How to unblock
CLIP (ablation configuration C)	**BLOCKED**	Allow huggingface.co in the cloud environment's network settings; code is ready.
Sentence-Transformers, SciBERT, NLI models	**BLOCKED**	Same — models are configured but cannot be downloaded.
Downloading data inside the cloud session	**BLOCKED**	Same; SciCap was fetched on a local machine instead.

7. Recommended Next Steps

- **1. Annotate ~100 SciCap figures** with element → entity links to measure grounding on real data.
- **2. Allow HuggingFace** to enable CLIP, Sentence-Transformers, SciBERT and NLI, then re-run all evaluations.
- **3. Replace the caption template** with a vision-language model (requires step 2).
- **4. Improve entity extraction** (SciBERT or a trained model) — 130 of 287 real figures currently get no entities.
- **5. Connect the Cross-Modal Search tab** and file upload in the UI to the backend.
- **6. Collect a small labelled set of real 3D figures** before making any claim about 3D on real papers.

8. How to Reproduce

Step	Command
Build synthetic benchmark	python -m src.data.benchmark_generator --out data/benchmark
Train / evaluate 3D models	python pipelines/evaluate_phase4.py
Train / evaluate grounding	python pipelines/evaluate_gnn.py
Ablation A–E	python pipelines/run_ablation.py
Real SciCap evaluation	python pipelines/evaluate_scicap.py
3D detector modes	python pipelines/evaluate_3d_detector_modes.py
Tests	python -m pytest tests/
Backend / UI	uvicorn api.main:app --port 8000 ; cd frontend && npm run dev
This report	python scripts/make_status_report.py

Hardware for the reported runs: Linux-6.18.44-fc-v114-x86_64-with-glibc2.39, PyTorch 2.14.1+cu130, CPU only.